Unit 1: Introduction (3 hrs)

Definition of microprocessor and its application

MICROPROCESSOR

A microprocessor (sometimes abbreviated µP) is a digital electronic component with miniaturized transistors on a single semiconductor integrated circuit (IC) .It is a multipurpose, Programmable clock-driven, register based electronic device that read binary instruction from a storage device called memory, accepts binary data as input and processes data according to those instructions and provides results as outputs. A Microprocessor is a clock driven semiconductor device consisting of electronic circuits manufactured by using either a LSI or VLSI technique.

A typical programmable machine can be represented with three components: MPU, Memory and I/O as shown in Figure

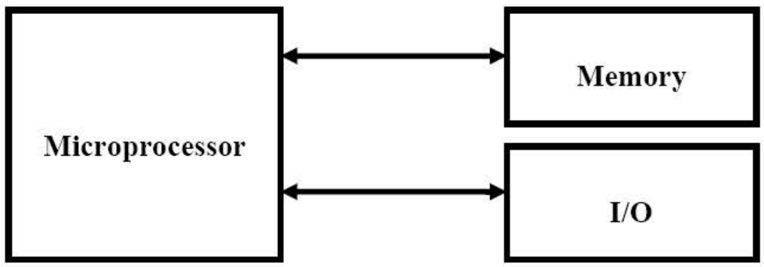


Figure: A Programmable Machine

These three components work together or interact with each other to perform a given task; thus they comprise a system. The machine (system) represented in above figure can be programmed to turn machine on and off, compute mathematical functions, or keep trace of guidance system. This system may be simple or sophisticated, depending on its applications.

Application of Microprocessor

The MPU applications are classified primarily in two categories: reprogrammable systems and embedded systems

Reprogrammable system: In this microprocessor is used for computing and data processing.

It is capable of handing large data, storage devices such as disks and CD Rom and peripherals devices such as printers. Example: microcomputer.

MICROCOMPUTER

The term *microcomputer* is generally synonymous with personal computer, or a computer that depends on a microprocessor. Microcomputers are designed to be used by individuals, whether in the form of PCs, workstations or notebook computers. A microcomputer contains a CPU on a microchip (the microprocessor), a memory system (typically ROM and RAM), a bus system and I/O ports, typically housed in a motherboard.

Microcomputers are small computers. They range from small controllers that work directly with 4-bit words to larger units that work directly with 32-bit words. Some of the more powerful Microcomputers have all or most of the features of earlier minicomputers. Examples of Microcomputers are IBM PC and Apple Macintosh computer.

Embedded system: In this case microprocessor is a part of final product and is not available for reprogramming to end uses. Example: washing machine, traffic light controller, Automatic testing machine. Here it is generally termed as microcontroller.

MICROCONTROLLER

It is a highly integrated chip that contains all the components comprising a controller. Single-chip Microcomputers are also known as Microcontrollers. They are used primarily to perform dedicated functions. They are used primarily to perform dedicated functions or as slaves in distributed processing.

Generally they include all the essential elements of a computer on a single chip: MPU,R/W memory, ROM and I/O lines and timers. Unlike a general-purpose computer, which also includes all of these components, a microcontroller is designed for a very specific task - to control a particular system.

A microcontroller is meant to be more self-contained and independent, and functions as a tiny, dedicated computer. The great advantage of microcontrollers, as opposed to using larger microprocessors, is that the parts-count and design costs of the item being controlled can be kept to a minimum. Microcontrollers are sometimes called embedded microcontrollers, which just mean that they are part of an embedded system that is, one part of a larger device or system. Typical examples of the single-chip microcomputers are the Intel 8051, AT89C51, AT89C52 and AVR,PIC

Both types of system can be used in various consumer applications like:

Transportation Industry

Automobiles, trains and planes utilize microprocessor technology. The 1978 Cadillac Seville was the first consumer vehicle to use a microprocessor embedded in its digital display. The "Trip Computer" provided mileage and additional information on the current trip.

Microprocessors work behind the scenes to keep commuters safe and timely. Consumer vehicles--cars, trucks, RVs-- integrate microprocessors to communicate important information throughout the vehicle. For example, navigation systems provide information using microprocessors and global positioning system (GPS) technology. In addition, microprocessors are used to control in-dash features like entertainment and climate control.

Mass transportation systems like flights and trains rely on microprocessors for important information. Public transportation fare cards, or smart-cards, contain processors to calculate fares, deduct the appropriate amount and retain information on how much funding remains. The aviation system relies heavily on microprocessors from calculating weather conditions to controlling the complex functions of an airplane.

Computers and Electronics

The "brain" of the computer, microprocessors drive technology. They are used in each type of computer from supercomputers to microcomputers. In addition, many electronic devices have central processing units (CPU) embedded. The CPU performs computer processing tasks by executing software instructions relative to the data it contains. For example, a cell phone or mobile device executes game instructions by way of the microprocessor. While playing chess, the microprocessor holds data about your last action and executes software instructions for the "computer's" next move. VCRs, televisions and gaming platforms also contain microprocessors for executing complex tasks and instructions.

Household Devices

A complex home security system or the programmable thermostat neatly attached to the wall contains microprocessor technology. Technology-based home security system microprocessors assist with monitoring large or small properties. The simplest programmable thermostat allows you to control the temperature in your home. Entering the preferred degree and achieving it on a consistent basis requires some intelligence on the part of the thermostat. A microprocessor in the system works with the temperature sensor to determine and adjust the temperature accordingly. Dishwashers, washing machines, high-end coffee makers and radio clocks contain microprocessor

Evolution of microprocessor

EVOLUTION OF INTEL SERIES MICROPROCESSOR

Intel 4004* Microprocessor Chip

In 1969, Intel Corporation began work on a project to develop a set of chips for a series of high-performance programmable calculators for Busicom, a Japanese company. Marcian E. "Ted" Hoff, who had joined Intel in 1968, is assigned to the project. Ted Hoff, along with Federico Faggin, Stan Mazor and others developed a design that included four chips.

The four chip combination included a central processing unit chip (CPU), a read-only memory chip (ROM), and a random access memory chip (RAM), and a shift-register chip for input and output (IO). This design was the first microprocessor chip, which Intel named the 4004.

The Intel 4004 was one-eighth of an inch wide by one-sixteenth of an inch long and contained 2300 metal-oxide semiconductor transistors (MOS). Its computing power was equal to the giant 18,000 vacuum tube ENIAC built in 1946.

The Intel 4004 could execute 60,000 operations per second. Masatoshi Shima, of Busicom, designed the logic for the chip. Shima later joined Intel. Intel sold Busicom the processor design for $60,000, but later bought back the design rights when Stan Mazor and Ted Hoff lobbied for the many other potential uses of the 4004 chip.

Intel 8008* Chip and the Intel 8080* Chip

The 8008 was an 8-bit microprocessor chip and was introduced in April 1972. Designers were Ted Hoff, Federico Faggin, Stan Mazor and Hal Feeney.

An even greater achievement was the 8080 chip, introduced in 1974. The 8080 had 10 times the performance of the 8008 chip and could execute 290,000 instructions per second. It had 64 bytes of addressable memory, and sold for $360 per chip. It quickly became an industry standard. Designers included Mazor, Faggin and Masatoshi Shima.

Intel 8086* Microprocessor

The 8086, announced by Intel Corporation in 1978, had 10 times the performance of the 8080 chip announced in 1974.

The 8086 established a new 16-bit software architecture. The project team included Bill Pohlman, Bob Koehler, John Bayliss, Jim Mckevitt, Chuck Wildman, Steve Morse and others. Motorola introduced the 68000 chip a year later, which directly competed with the 8086. By 1984, however, the 8086 chip was outselling the 68000 by approximately 9 to 1. The 8088 chip was released in 1981.

Intel 80286* Microprocessor

In 1982, Intel Corporation released the 80286 microprocessor chip. At the time of its introduction, the 80286 microprocessor has three times the performance of any other 16-bit chip on the market. The 80286 offered on-chip memory management, making it suitable for multitasking operations. Intel's project leader for the 286 is Gene Hill. Intel also released the 80186 chip, which was an improvement over earlier Intel chips. The 80186 design team was lead by Dave Stamm.

Intel 80386* Microprocessor chip

The Intel 80386 is a 32-bit microprocessor containing over 275,000 transistors on a single chip. The 80386 (commonly known as the "386 chip") could handle four million operations per second and handle memory up to four gigabytes (4,294,967,296). The 386 was also compatible with Intel's earlier processor line for the IBM PC and compatibles and could run software designed for those processors as well. The 386 chip brought desktop personal computing power to a new level. John Crawford was the architecture manager for the Intel 386 and the Intel 486 microprocessors, and co-manager of Pentium microprocessor development.

Intel 80486* Microprocessor chip

In 1989, Intel Corporation announced the 80486 chip, a highly integrated 32-bit microprocessor combining 80386 compatibility, RISC-style CPU, 80387 math co-processor compatibility, 8-Kilobyte on-chip cache and built- in multiprocessing support. The 80486 has a reported capability of holding 1.16 million transistors and is about four times faster than the 80386 processor. Initial uses of the 80486 chip will be for LAN servers and high- end workstations.

John Crawford was the architecture manager for the Intel 386 and the Intel 486 microprocessors, and co-manager of the Pentium microprocessor development. The most common varieties of the 80486 chip are the 486SX (25Mhz), 486DX (33Mhz), and the 486DX2 (66Mhz).

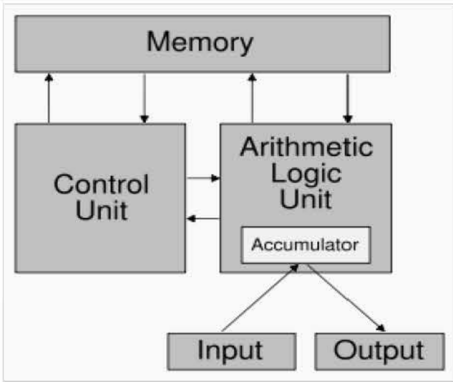
Intel Pentium Microprocessor

In 1993, Intel announced the Pentium chip. The word "Pentium" comes from the Greek root word "pentas" meaning "five." The Pentium is the 80586 chip.

The Pentium is a 32-bit chip with superscalar design, and is estimated to be two times faster than the 486DX2 (66MHz) chip. The Pentium uses dual pipelines to allow it to process two separate instructions in a single cycle. The Pentium has a 64 bit bus interface, an eight bit code cache, an eight bit data cache, and branch prediction memory bank. Don Alpert was the architecture manager of the Pentium, John Crawford was co-manager. The Pentium is a CISC-based (complex instruction set computer) chip containing 3.3 million transistors.

Von Neumann architecture

The task of entering and altering the programs for the one of early computer called ENIAC (electronic numerical integrator and computer) was extremely tedious. The programming concept could be faciliated if the program could represented in a form suitable for storing in memory along side the data.Than a computer could get it's instruction by reading them form the memory and a program could be set or altered by setting the values of a portion of memory . This approach is known as **stored program concept**.



Design of the von Neumann architecture

The von Neumann architecture is a computer design model that uses a processing unit and a single separate storage structure to hold both instructions and data. It is named after mathematician and early computer scientist John von Neumann. The term "stored-program computer" is generally used to mean a computer of this design. The von Neumann architecture allows for one instruction to be read from memory or data to be read/written from/to memory at a time. Instructions and data are stored in the same memory subsystem and share a communication pathway or bus to the CPU.

A stored-program design also lets programs modify themselves while running, effectively allowing the computer to program itself. One early motivation for such a facility was the need for a program to increment or otherwise modify the address portion of instructions, which had to be done manually in early designs. This became less important when index registers and indirect addressing became customary features of machine architecture. Self-modifying code has largely fallen out of favor, since it is very hard to understand and debug, as well as inefficient under modern processor pipelining and caching schemes.

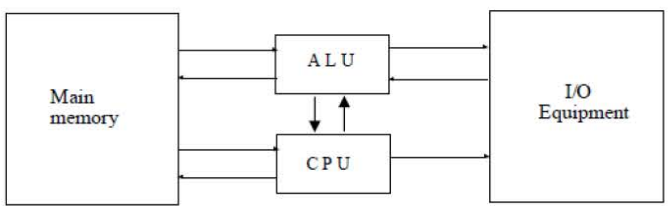


Fig: Von-Numann Architecture.

The memory of Von-Neumann machine consists of thousand storage location called words of 40 binary digits(bits). Both data and instruction are stored in it.

The storage locations of control unit and ALU are called registers. The various registers of this model are MBR, MAR, IR, IBR, PC, AC.

Memory Buffer Register: It consist of a word to be stored in memory or is used to receives a memory or is used to receive a word from memory.

Memory address Register: It contain the address in memory of the world to be written from or read into the MBR.

IR(Instruction register): Contain the 8 bit upcode (operation code) instruction being executed.

IBR(instruction buffer register): It is used to temporarily hold the instruction from a word in memory.

PC (program counter): It contain address of next instruction to be fetched from memory.

Ac (Accumulator) and MQ(multiplier quotient): They are employed to temporarily hold operands and results of ALU operations.

The separation between the CPU and memory leads to the von Neumann bottleneck, the limited throughput (data transfer rate) between the CPU and memory compared to the amount of memory. In modern machines, throughput is much smaller than the rate at which the CPU can work. This seriously limits the effective processing speed when the CPU is required to perform minimal processing on large amounts of data. The CPU is continuously forced to wait for vital data to be transferred to or from memory. As CPU speed and memory size have increased much faster than the throughput between them, the bottleneck has become more of a problem.

Basic organization of microcomputer (system using microprocessor)

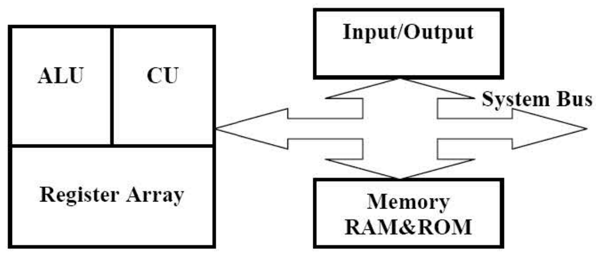


Figure: Microprocessor Based System with Bus Architecture.

Figure shows a block diagram for a simple Microcomputer. The major parts are the CPU, Memory and I/O. These three parts are connected by three sets of parallel lines called buses. The three buses are address bus, data bus and the control bus.

MEMORY

The First Purpose of memory is to store binary codes for the sequences of instructions you want the computer to carry out. It consists of RAM and ROM.The second purpose of the memory is to store the binary-coded data with which the computer is going to be working.

INPUT/OUTPUT

The input/output or I/O Section allows the computer to take in data from the outside world or send data to the outside world. Peripherals such as keyboards, video display terminals, printers are connected to I/O Port.

CPU

The CPU controls the operation of the computer. In a microcomputer CPU is a microprocessor. it fetches binary coded instructions from memory, decodes the instructions into a series of simple actions and carries out these actions in a sequence of steps. The CPU also contains an address counter or instruction pointer register, which holds the address of the next instruction or data item to be fetched from memory.

COMPONENTS OF CPU (Microprocessor)

The CPU is divided into three segments: ALU, Register array and Control Unit.

ARITHMETIC LOGIC UNIT

This is the area of Microprocessor where various computing functions are performed on data. The ALU performs operations such as addition, subtraction and logic operations such as AND, OR and exclusive OR.

REGISTER ARRAY

These are storage devices to store data temporarily. There are different types of registers depending upon the Microprocessors. These registers are primarily used to store data temporarily during the execution of a program and are accessible to the user through the instructions. General purpose Registers of 8086 includes AL, AH, BL, BH, CL, CH, DL, DH. There is also instruction pointer and decoder to decode the instructions fetched from memory.

CONTROL UNIT

The Control Unit Provides the necessary timing and control signals to all the operations in the Microcomputer. It controls the flow of data between the Microprocessor and Memory and Peripherals. The Control unit performs 2 basic tasks-

1. SEQUENCING The control unit causes the processor to step through a series of micro-operations in the proper sequence, based on the program being executed. It decodes instruction to generate it.

2. EXECUTION The control unit causes each micro operation to be performed.

CONTROL SIGNALS

For the control unit to perform its function it must have inputs that allow it to determine the state of the system and outputs that allow it to control the behavior of the system.

•Inputs:

Clock, Instruction Register, Flags

•Outputs:

Control signals to Memory
Control signals to I/O
Control Signals within the Processor.

SYSTEM BUS

Typical system uses a number of busses, collection of wires, which transmit binary numbers, one bit per wire. A typical microprocessor communicates with memory and other devices (input and output) using three busses: Address Bus, Data Bus and Control Bus.

ADDRESS BUS

The address bus consists of 16, 20, 24 or 32 parallel signal lines. On these lines the CPU sends out the address of the memory location that is to be written to or read from. The no of memory location that the CPU can address is determined by the number of address lines. If the CPU has N address lines, then it can directly address 2^n memory locations, i.e. CPU with 16 address lines can address 2^{16} or 65536 memory locations.

DATA BUS

The data bus consists of 8, 16 or 32 parallel signal lines. The data bus lines are bi-directional. This means that the CPU can read data in from memory or it can send data out to memory

CONTROL BUS

The control bus consists of 4 to 10 parallel signal lines. The CPU sends out signals on the control bus to enable the output of addressed memory devices or port devices. Typical control bus signals are Memory Read, Memory Write, I/O Read and I/O Write.